

Literature Status: Diffeological Regularity and an ILB Frobenius Theorem

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Status: literature already answered. The regularity/Frobenius question in arXiv:1710.11076 is explicitly answered by the same author's later revision arXiv:1607.02636v3. Cauchy diffeology and the subset diffeology make the implicit map smooth on a suitable non-open domain, and Theorem 4.3 proves the corresponding Frobenius theorem.

1 Original question

J.-P. Magnot's *On the domain of implicit functions in a projective limit setting without additional norm estimates*, arXiv:1710.11076 [1], constructs an implicit map

$$u_\infty : D_\infty \longrightarrow F_\infty$$

on a domain that may fail to be open in the projective-limit topology. The domain nevertheless contains the unit ball $\mathcal{B}$ of an auxiliary Banach space continuously embedded in the projective limit.

Section 2.2, arXiv PDF page 6, reaches the point where the usual proof of the Frobenius theorem needs D_1J on D_∞ and asks:

Source open problem. *Classical differentiation applies on open domains, whereas D_∞ need not be open. Among the generalized notions of differentiation, which is best adapted to this setting?*

The abstract and introduction identify the same issue as the missing regularity framework needed for a projective-limit Frobenius theorem.

2 Separate later answer

The same author's *On the differential geometry of numerical schemes and weak solutions of functional equations*, arXiv:1607.02636v3 [2], is the separate answering work. The numbering of the arXiv identifiers is potentially misleading: v3 was revised on 12 August 2020, after arXiv:1710.11076, and is the version accepted for publication in *Nonlinearity*.

On PDF page 20 the later paper restates the earlier theorem, says explicitly that arXiv:1710.11076 left regularity open, and announces that it fills the gap using Cauchy diffeology. Its Theorem 4.2 on PDF page 21 proves:

Later answer. *There exists a domain D with*

$$\mathcal{B} \subset D \subset D_\infty$$

such that the implicit function u_∞ is smooth when D carries its subset diffeology.

The proof realizes u_∞ as the limit of the contraction iterates $\phi_{i,x}^n(0)$ at every Banach level. The parameter-to-iterate map is smooth into the space of Cauchy sequences equipped with the Cauchy diffeology, and the limit map is smooth by construction. Passing to the projective intersection gives the asserted smooth map on D .

Theorem 4.3, beginning on PDF page 22 and continuing on page 23, then applies this result to the path-space implicit equation used in the source. Under the levelwise ILB Frobenius compatibility condition, it produces a diffeological subspace D and a smooth map $J : D \rightarrow F_\infty$ satisfying

$$D_1 J(x, y) = f_i(x, J(x, y))$$

together with the required normalization on the connected fiber over the base point. This is the Frobenius theorem whose classical proof was blocked in the source.

Scope

The later paper selects a precise generalized calculus, proves regularity in that calculus, and derives the desired Frobenius statement. It therefore answers the source’s regularity/Frobenius open problem rather than merely suggesting terminology.

The source has a second, separate closing question about the structure of $GL_\infty/GL(B)$. The cited theorems do not classify that quotient, and this packet makes no claim about it. No new mathematical result is claimed here; this packet records an author-explicit later-literature answer.

References

- [1] J.-P. Magnot, *On the domain of implicit functions in a projective limit setting without additional norm estimates*, Demonstr. Math. **53** (2020), 112–120; arXiv:1710.11076.
- [2] J.-P. Magnot, *On the differential geometry of numerical schemes and weak solutions of functional equations*, Nonlinearity **33** (2020), 6835–6867; doi:10.1088/1361-6544/abaa9f; arXiv:1607.02636v3.